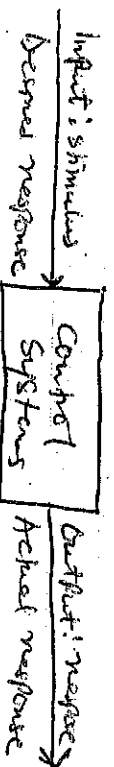


CONTROL ENGINEERING I

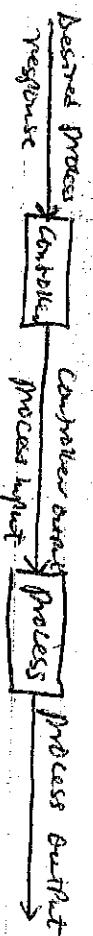
⇒ Definition of Control Systems: A Control System consists of subsystems and processes assembled for the purpose of obtaining a desired output with desired performance, given a specific input.



Or a control system is a system of devices or set of devices that manages, commands, directs or regulates the behavior of other devices to achieve desired results.

⇒ OPEN LOOP CONTROL SYSTEMS

A control system in which the control action is totally independent of the output of the system. It is called open loop control system as shown below.



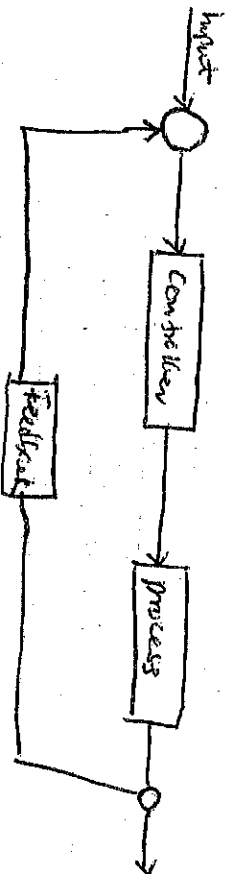
Examples of open loop control systems:

1) Electric hand drier - Hot air (output) comes out as long as you keep your hand under the machine irrespective of how much your hand is dried.

2) Automatic Washing Machine - This machine runs according to the preset time irrespective of washing is completed or not.

3) Light switch - Lamp glow whenever the light switch is on irrespective of light is required or not.

Closed Loop Control Systems: Control systems in which the output has an effect on the input quantity in such a manner that the input quantity will adjust itself based on the output generated is called a closed-loop control system. An open loop control system can be converted into a closed loop control system by providing feedback. This feedback automatically makes suitable changes in the output due to external disturbance as shown below



Examples of Closed loop control systems:

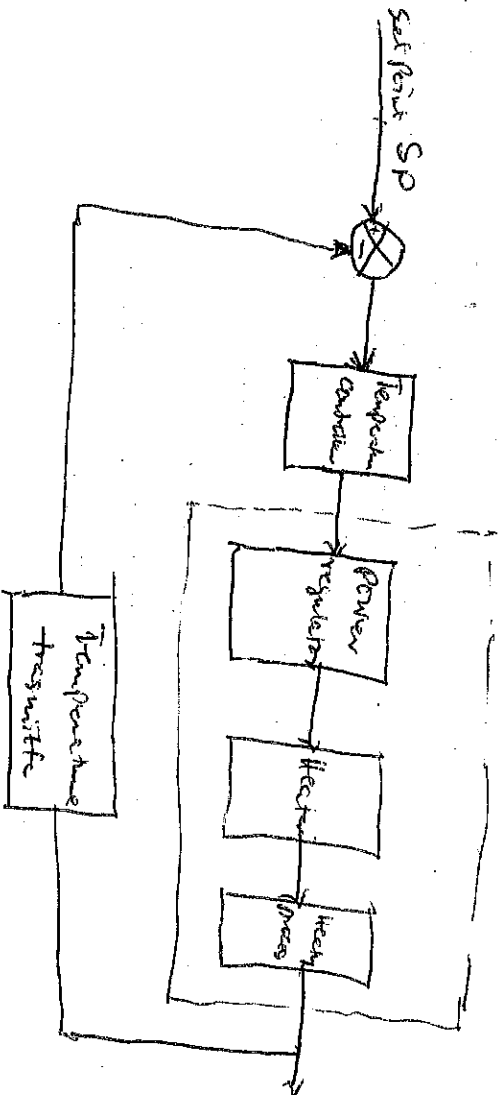
- 1) Automatic Fluorescent room-heating elements are controlled by the output temperature of the room.
- 2) Voltage stabilizer: Voltage controller operates depending upon the output voltage of the system.
- 3) Water level controller: Input water is controlled by the water level of the reservoir.

FEED BACK LOOP

The feedback loop is the feed-back that takes the system output into consideration and tunes the system to adjust its performance to meet the desired need of the system.

= SYSTEM:

Block diagram of water heater control



Block diagram of the control system of Refrigeration

